

# Jinghang Li

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## Summary

PhD candidate in Bioengineering focused on medical imaging AI (MRI) and computer vision. Built end-to-end pipelines across data curation, preprocessing, multi-GPU training (DDP), and deployment on HPC/Slurm and GPU servers. Experienced in generative modeling (diffusion, flow matching), segmentation, cross-modal feature matching, and physics-informed optimization (RF shimming). Motivated by translating research prototypes into robust, fast inference workflows.

## Education

|            |                                                                                     |                    |
|------------|-------------------------------------------------------------------------------------|--------------------|
| <b>PhD</b> | <b>University of Pittsburgh</b> , Bioengineering – Pittsburgh, PA, USA              | Aug 2021 – present |
|            | <b>Carnegie Mellon University</b> , Visiting Graduate Student – Pittsburgh, PA, USA | 2021 – 2023        |
| <b>BS</b>  | <b>University of Pittsburgh</b> , Bioengineering – Pittsburgh, PA, USA              | 2016 – 2021        |

## Awards

- Big Idea Competition — 3rd Place, University of Pittsburgh (April 2026)
- Department Best Teaching Assistant (Nominated), University of Pittsburgh (2023)

## Skills

- Languages:** Python, MATLAB, R, C, Bash, LaTeX
- ML / Computer Vision:** PyTorch, MONAI, TorchIO, torchvision, U-Net, Transformers, Diffusion, Flow Matching, VAE
- Medical Imaging:** MRI, GRAPPA, DICOM/NIfTI, preprocessing (registration, skull stripping, normalization), segmentation
- Compute & MLOps:** Linux, HPC/Slurm, multi-GPU DDP, Docker, Singularity, Git

## Projects

- Flow Matching for Single-Step MR Image Enhancement – Pittsburgh, PA, USA** Jan 2025 – present  
Led a zero→one, clinically focused pipeline for single-step MRI enhancement—from training dataset curation to on-prem GPU deployment.
  - Built a paired 7T dataset and preprocessing toolkit (crop/pad, intensity normalization, skull stripping, registration); scaled processing via Slurm job arrays on HPC.
  - Trained conditional diffusion and flow-matching models in PyTorch/MONAI; enabled multi-GPU DDP training on HPC for faster iteration.
  - Reduced inference latency by progressing from diffusion acceleration (DDIM, DeepCache-style caching) to conditional flow matching built on MONAI diffusion UNet codebase.
  - Productionized inference for NIfTI/DICOM with Docker/Singularity; deployed on a 4U GPU server (2× RTX 6000 Ada) integrated with the MRI scanner workflow. [VERIFY integration details]
  - Submitted invention disclosure to the university innovation office; provisional patent filed.
- GPU-Accelerated RF Shimming Optimization (60-channel Tx) – Pittsburgh, PA, USA** Jan 2022 – Dec 2023  
Physics-informed optimization for large-scale RF shim parameter search under scanner power constraints.
  - Implemented RF shimming optimization in PyTorch to leverage GPUs and Slurm for large-scale optimizations.
  - Optimized amplitudes/phases across 60 transmit channels over large FOV volumes (~320×320×400)
  - Implemented differentiable histogram objective function from literature to improve solution quality vs prior baselines.

## Publications

- wmh\_seg: Transformer based U-Net for Robust and Automatic White Matter Hyperintensity Segmentation across 1.5 T, 3T and 7T.** Feb 2024  
*Jinghang Li*, Tales Santini, Yuanzhe Huang, et al., Minjie Wu  
[arxiv.org/abs/2402.12701](https://arxiv.org/abs/2402.12701) (arxiv)

**Bridging Neuroimaging and Neuropathology: A Comprehensive Workflow for Targeted Sampling of White Matter Lesions**

Oct 2025

*Jinghang Li*, Nadim Farhat, Jacob Berardinelli, et al.,

<https://doi.org/10.1111/jon.70094> (Neuroimaging)

**MRecover: A Conditional Generative Model for Recovering Motion-Corrupted MR Images Using AI Generated Contrast**

May 2026

*Jinghang Li*, Tales Santini, Courtney Clark, et al., Minjie Wu, Tamer S. Ibrahim

[arxiv.org/abs/2605.21669](https://arxiv.org/abs/2605.21669) (arxiv)